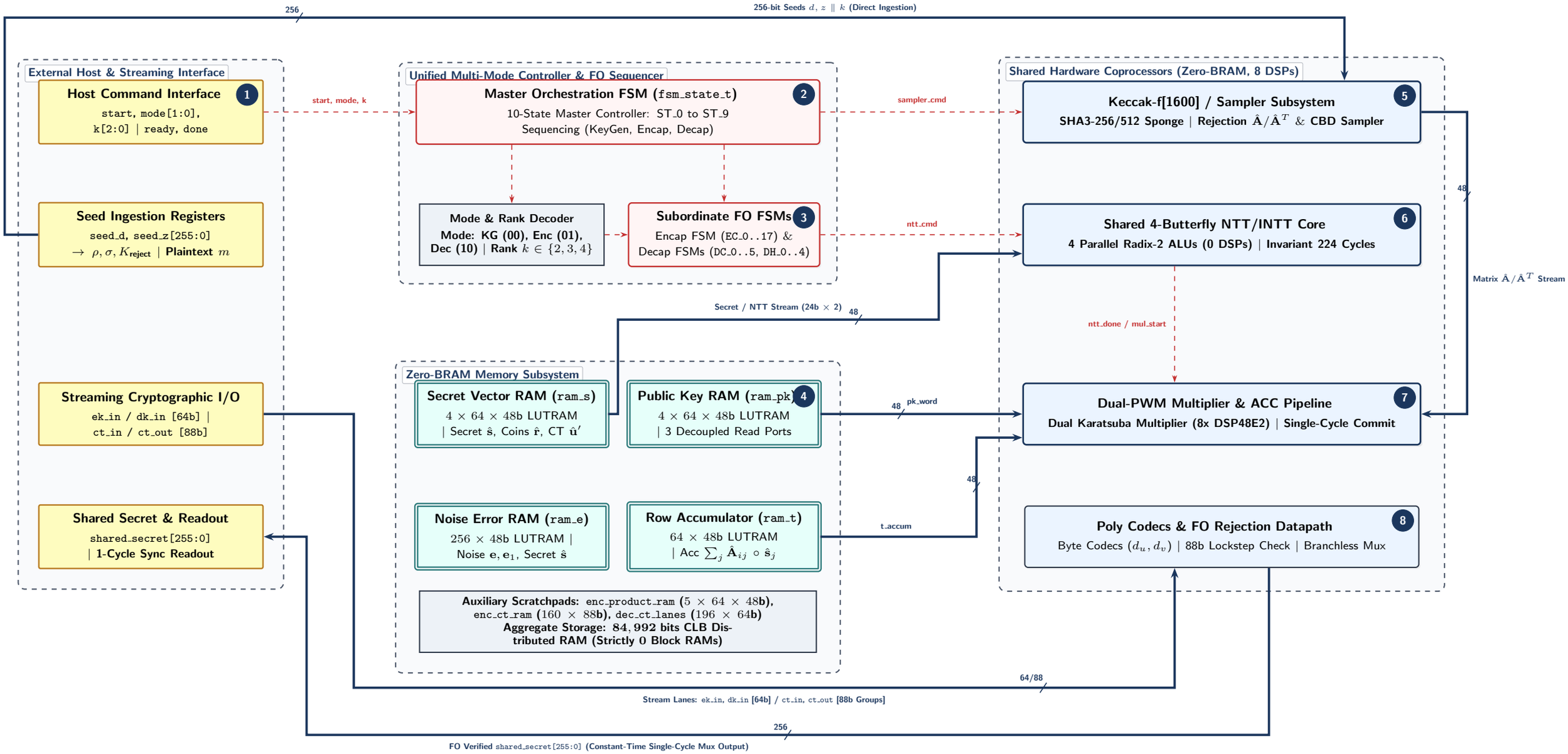
# ML-KEM HPKA: Top-Level System Architecture, Multi-Mode Controller & FO-Transform Datapath

Subsystem 01 Microarchitecture — Unified FIPS 203 KeyGen / Encapsulation / Decapsulation Core — RTL: `mlkem_keygen_top.sv`




| ID | Component / Signal | Width / Format | Description |
|---|---|---|---|
| **1** | Host Command Interface | Control / Handshake | Latches start trigger, operating mode (KeyGen, Encap, Decap), and rank $k$; exposes status flags and cycle profiler. |
| **2** | Master Orchestration FSM | 10-State Master FSM | Top-level controller executing states ST_0 through ST_9; manages mode branching and synchronization across all engines. |
| **3** | Subordinate FO Controllers | Hierarchical FSMs | Encap datapath FSM (EC_0..17) and Decap FSMs (DC_0..5, DH_0..4) driving PKE encryption, message recovery, and re-encryption. |
| **4** | Zero-BRAM Distributed RAM | 84,992b LUTRAM | CLB Distributed RAM storing secret vectors, public keys, noise polynomials, and intermediate row sums with strictly 0 BRAMs. |
| **5** | Keccak-f[1600] / Sampler | 64b Lane Streaming | Shared cryptographic sponge computing SHA3-256/512 digests, SHAKE-128 matrix sampling, and SHAKE-256 CBD error sampling. |
| **6** | 4-Butterfly NTT/INTT Core | 4x Radix-2 ALUs (0 DSP) | Soft-logic transform engine computing forward NTT and inverse INTT in an invariant 224 cycles per polynomial. |
| **7** | Dual-PWM Multiplier & ACC | 8x DSP48E2 Slices | Pipelined Karatsuba multiplier and 4-stage modular accumulator computing $\hat{\mathbf{A}} \circ \hat{\mathbf{s}} + \hat{\mathbf{e}}$ with single-cycle row commits. |
| **8** | Poly Codecs & FO Logic | Codecs + CT Mux | Handles coefficient compression/decompression, lockstep ciphertext comparison, and single-cycle branchless implicit rejection. |